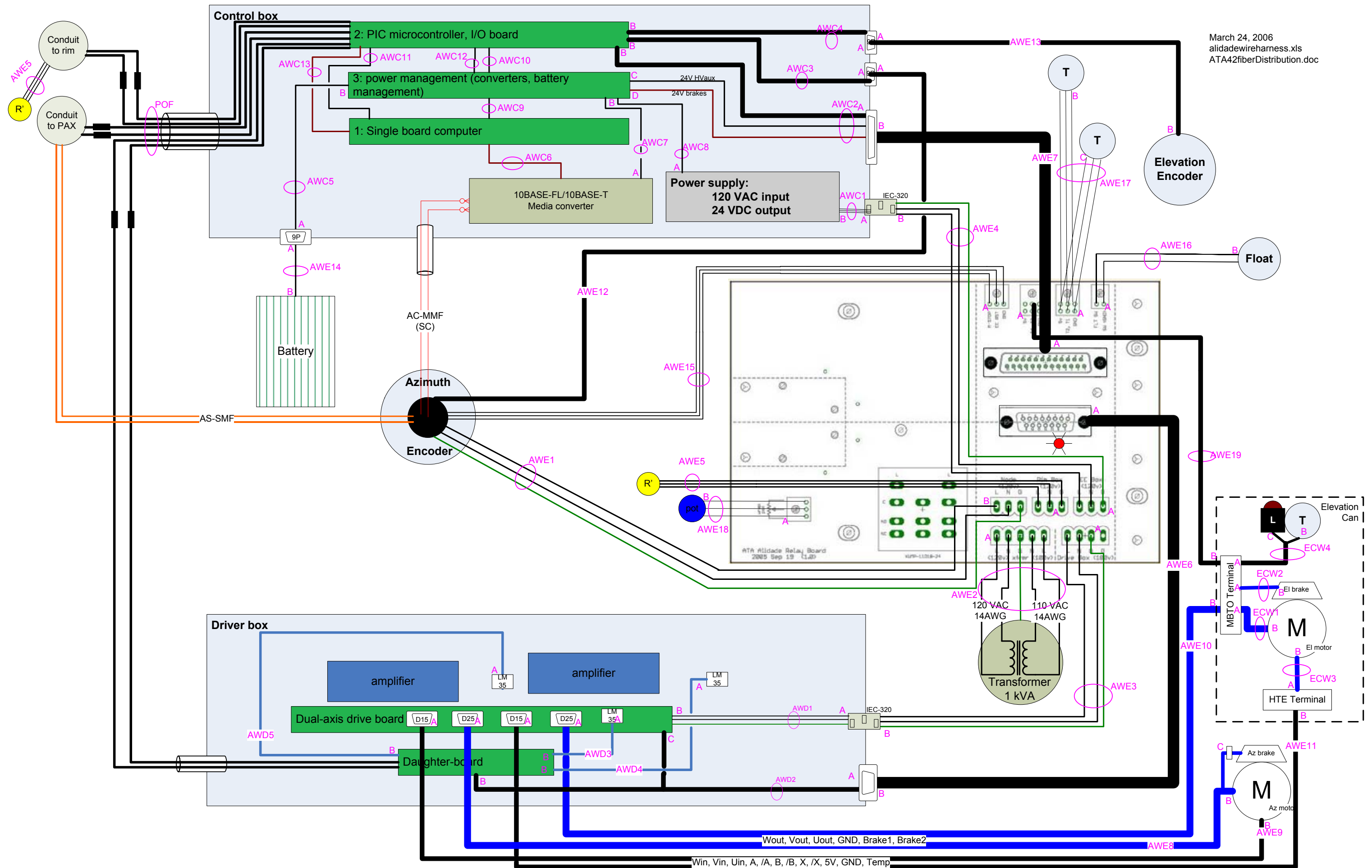




March 24, 2006
alidawireharness.xls
ATA42fiberDistribution.doc

Summary of Alidade Wiring

3/31/2006

Designator	Description	A/C connector	A/C Part Number	B/D connector	B/D Part Number	Length	Cable
AWE 1	3-wire input power from node	pie-plate terminal	---	plug on relay board	39860-0103		17' Belden 8527 (white/black) / 8520 (green)
AWE 2	5-wire for transformer/relay board	plug on relay board	39860-0105	---	---		CTF red part of transformer
AWE 3	3-wire for transformer/driver box	plug on relay board	39860-0104	plug on driver box	IEC-320		34 Volex 17539
AWE 4	3-wire power for control box	plug on relay board	39860-0103	plug on controller box	IEC-320		23 Volex 17539
AWE 5	3-wire power for rim box	plug on relay board	39860-0103	in rim box	---		#14 wire white/black/green
AWE 6	15-wire relay board/driver box	15-pin D to relay board	D15P	15-socket D to driver	D15S		-- Cables to Go
AWE 7	25-wire relay board/control box	25-socket D to relay bd	D25S	25-pin D to controller	D25P		-- Cables to Go (change)
AWE 8	motor/brakes for az	25-pin D to driver bd	D25P	motor	---		part of motor
		brake	39-01-3023-P/39-00-0041-C				
AWE 9	encoder/halls for az	15-pin D to driver bd	D15P	motor	---		part of motor
AWE 10	extender for motor/brakes for el	25-pin D to driver bd	D25P	MBTO terminal	---		158 Belden 83352 + #18 hook-up
AWE 11	extender for encoder/halls for el	15-pin D to driver bd	D15P	HTE terminal	---		158 Belden 8307
AWE 12	encoder for az	9-pin D on controller box	D9P	encoder	---		30 part of encoder
AWE 13	encoder for el	9-pin D on controller box	D9P	encoder	---		CTF part of encoder
AWE 14	battery for control box	spade on battery	---	9-socket D on cntrler	D9P		22/19 black/red wire
AWE 15	3-wire relay board to pie plate	micro-fit on relay bd	43645-0300/43030-009	pie-plate terminal	---		17' LappUSA (Olflex) 900P Cable, 3 conductor #24, 811442
AWE 16	2-wire for az oil float	micro-fit on relay bd	43645-0200/43030-009	float switch	---		-- part of switch
AWE 17	LM35 in alidade x 2	micro-fit on relay bd	43025-06000/43030-009	LM35	---		wire
AWE 18	3-wire az pot to relay board	micro-fit on relay bd	43645-0300/43030-009	pot	---		wire
AWE 19	6-wire from el can for oil and temp	micro-fit on relay bd	43025-06000/43030-009	MBTO terminal	---		158 Belden 9611
ECW 1	motors for elevation	MBTO terminal block	---	motor	---		cable from motor
ECW 2	brakes for elevation	MBTO terminal block	---	brake	39-01-3023-P/39-00-0041-C		cable from brake
ECW 3	encoder/halls for elevation	HTE terminal block	---	motor	---		cable from motor
ECW 4	temperature/oil indicator for elevation	MBTO terminal block	---	leads on device	---		wire
AWD 1	3-wire input power from relay board	terminals on plug	---	lug	---		
AWD 2	15-wire from AWE6 to driver/daughter boards	filtered D on cover	56-111-010	1x5-100 on daughter	50-57-9005/16-02-0102		
		terminal on driver board	---	lug	---		
AWD 3	3-wire temperature sensor on driver board	header on driver board	22-01-2037/08-52-123	1x3-100 on daughter	50-57-9003/16-02-0102		
AWD 4	3-wire temperature sensor on heat sink	device	---	1x3-100 on daughter	50-57-9003/16-02-0102		
AWD 5	3-wire temperature sensor in airflow	device	---	1x3-100 on daughter	50-57-9003/16-02-0102		
AWD 6	rectifier to capacitor						
AWD 7	capacitor 1 to capacitor 2						
AWD 8	resistors (6) to driver board						
AWD 9	resistors (6) to rectifier						
AWC 1	3-wire input power from relay board	terminal on plug	---	spade on power supply	---		
AWC 2	25-wire cable from AWE7	filtered D socket on cover	56-126-004	2x10-100 on cntrler bd	22-55-2201/16-02-0102		
		relay/brakes on pwr mgmt t	770849-2/770522-1	HV aux on pwr mgmt bd	770849-2/770522-1		
AWC 3	9-wire az encoder cable from AWE12A	filtered D socket on cover	56-106-010	2x5-100 on cntrler bd	22-55-2101/16-02-0102		
AWC 4	9-wire el encoder cable from AWE13A	filtered D socket on cover	56-106-010	2x5-100 on cntrler bd	22-55-2101/16-02-0102		
AWC 5	9-wire battery cable from AWE14A	filtered D pin on cover	56-101-010	pwr mgmt bd	770849-2/770522-1		
AWC 6	standard cat 5 cable						
AWC 7	9V to media converter from PMB	round plug	---	power mgmt bd	770849-2/770522-1		
AWC 8	24V to PMB from power supply	spade on power supply	---	power mgmt bd	770849-2/770522-1		
AWC 9	5V to SBC from PMB	plug on SBC	15-24-4048	power mgmt bd	770849-2/770522-1		
AWC 10	12V to cntrlr bd from PMB	cntrler board	770849-2/770522-1	power mgmt bd	770849-2/770522-1		
AWC 11	SBC reset	cntrler board	50-57-9005/16-02-0102	SBC	50-57-9005/16-02-0102		
AWC 12	electronics on/off	cntrler board	50-57-9003/16-02-0102	pwr mgmt bd	50-57-9003/16-02-0102		
AWC 13	cross-over cat 5 cable						

alidade enclosure (AWE)

controller box (AWC)

driver box (AWD)

elevation can (ECW)

Connector Summary

4/7/2006

Qty	Part Number	Location	Tooling
Molex Micro-fit 3.0 (0.118")			
2	43645-0300/43030-0009	AWE15A, AWE18A	Crimper: 63811-2800 Extractor: 11-03-0043
1	43645-0200/43030-0009	AWE16A	
2	43025-06000/43030-0009	AWE17A, AWE19A	
Molex Eurostyle 5.08 (0.200")			
1	39860-0105	AWE2A	Terminal block
1	39860-0104	AWE3A	
3	39860-0103	AWE1B, AWE4A, AWE5A	
Molex Mini-fit Jr. 4.20 (0.165")			
2	39-01-3023/39-00-0041	AWE8C, ECW2B	Crimper: 11-01-0197 Extractor: 11-03-0044
2	39-01-2020/39-00-0039	AWE8C, ECW2B	
1	39-01-4031/39-00-0066	ECW4	
1	39-01-4032/39-00-0041	ECW4	
Molex 0.100" SL crimp housing			
3	50-57-9005/16-02-0102	AWD2B, AWC11A, AWC11B	Crimper: 11-01-0208 Extractor: 11-03-0036
6	50-57-9003/16-02-0102	AWD3B, AWD4B, AWD5B, AWC12A, AWC12B, AWD3A	
1	22-55-2201/16-02-0102	AWC2B	
2	22-55-2101/16-02-0102	AWC3B, AWC4B	
Molex 0.200" disc drive power connect			
1	15-24-4048/02-08-1202	AWC9A	Crimper: 11-01-0206 Extractor: 11-03-0037
Tyco AMP SL-156 (MTA-156)			
8	770849-2/770522-1	AWC2C, AWC2D, AWC5B, AWC7B, AWC8B, AWC9B, AWC10A, AWC10B	Crimper: 90759-1 Extractor:
Spectrum Control Series 100 filtered D connectors			
1	56-111-010	AWD2A	
1	56-126-004	AWC2A	
2	56-106-010	AWC3A, AWC4A	
1	56-101-010	AWC5A	
RJ-45			
4	Cat 5 thru/crossover	AWC6/AWC13	
MBTO/HTE terminal blocks			
2	Wago 260-301x30/260-361x2	ECW1-4A, AWE19B, AWE10B, AWE11B	Connex